

**S226 PRT661 DATA SCIENCE PRACTICE****Assessment 2 Group Project Progress Report
Danala Group 8 Theme 2****Electricity Demand and Price Forecasting for the Australian National Electricity Market****Group Members**

1. DIPESH SEDHAI	S395457	S395457@students.cdu.edu.au
2. DILLIRAM DHAKAL	S396633	S396633@students.cdu.edu.au
3.NITESH RAJ CHAUDHARY*	S394811	S394811@students.cdu.edu.au
4.ROSHAN KUMAR SHRESTHA	S395498	S395498@students.cdu.edu.au

GitHub repository:

<https://github.com/niteshS394811/S226-PRT661-DATA-SCIENCE-PRACTICE>

Jira project board:

<https://shrestharoshan931-1785816864381.atlassian.net/jira/software/projects/KAN/list>

September 2026

Section 1 – Assessment 1 Report (Original Proposal)

The following is the Assessment 1 Group Project Proposal content as submitted by Danala Group 8 (Theme 2). It is included here to satisfy the Assessment 2 requirement that the complete Assessment 1 report form part of the single consolidated submission.

**S226 PRT661 DATA SCIENCE PRACTICE****Assessment 1 Group Project Proposal
Danala Group 8 Theme 2****Electricity Demand and Price Forecasting for the Australian National Electricity Market****Group Members**

1. DIPESH SEDHAI	S395457	S395457@students.cdu.edu.au
2. DILLIRAM DHAKAL	S396633	S396633@students.cdu.edu.au
3.NITESH RAJ CHAUDHARY*	S394811	S394811@students.cdu.edu.au
4.ROSHAN KUMAR SHRESTHA	S395498	S395498@students.cdu.edu.au

1. Project Overview and Objectives

This project aims to develop an end-to-end predictive analytics system that will offer forecasts of regional electricity demand and wholesale electricity prices within the 5 National Electricity Market (NEM) regions (QLD1, NSW1, VIC1, SA1 and TAS1). The system will be based on historical and the latest data released by the Australian Energy Market Operator (AEMO) via NEMWEB. The integration of demand, price and generation data will be connected with the common time-series data of rooftop PV and interconnector-flow.

The above mentioned pipeline is proposed to include data acquisition, ETL processing, storage, feature engineering, model development, evaluation, forecasting and visualisation. Automated workflow orchestration will be done using Apache Airflow, local relational storage will be enabled by SQL Server, the development of the LSTM/GRU model will be done using TensorFlow and the final interactive visualization will be done using Tableau Public.

The project has 4 key objectives:

- Make short-term multi-horizon predictions of regional electricity demand and Regional Reference Price (RRP).
- Include more explanatory variables, such as generation mix, rooftop PV and interconnector flows.
- Set up an automated Airflow pipeline for data ingestion, transformation, storage and forecast generation.
- Create Tableau Public dashboards of actual vs. forecast values, regional comparisons, and model performance.

2. Theme Justification

The project fits well within the scope of Theme 2 – Predictive Analytics and Forecasting as its main goal is to predict the future electricity demand and wholesale prices, given past observations and explanatory variables. The proposed workflow consists of data acquisition, data preparation, time-series feature engineering, model development, evaluation and visualization, which is directly related to the requirements of the chosen theme.

The LSTM and GRU models will be explored since they are specifically created for sequential data and learn temporal dependencies (Hochreiter & Schmidhuber, 1997; Cho et al., 2014). Feature construction will be based on established forecasting principles, as will the chronological validation and evaluation of the forecasts (Hyndman & Athanasopoulos, 2021).

3. Problem Statement

The utility of short term forecasts for monitoring and decision making in the NEM is important. While AEMO offers well established market forecasts, this project will explore the possibility of making improvements to this forecasting through the use of the integration of demand, price, generation, rooftop PV and interconnector characteristics with machine learning models.

The project will create an independent forecasting system based on the data from NEMWEB for the five NEM regions. Observations will be cleaned and merged, and then converted to time-series features like lagged demand and price, rolling statistics and calendar variables. LSTM/GRU models will produce multi-horizon forecasts for the demand and RRP.

The project will evaluate the feasibility and predictive performance of this feature-based automated forecasting approach against relevant forecasting baselines.

4. Initial Architecture Diagram

The system will follow a layered architecture, orchestrated end-to-end by Apache Airflow:

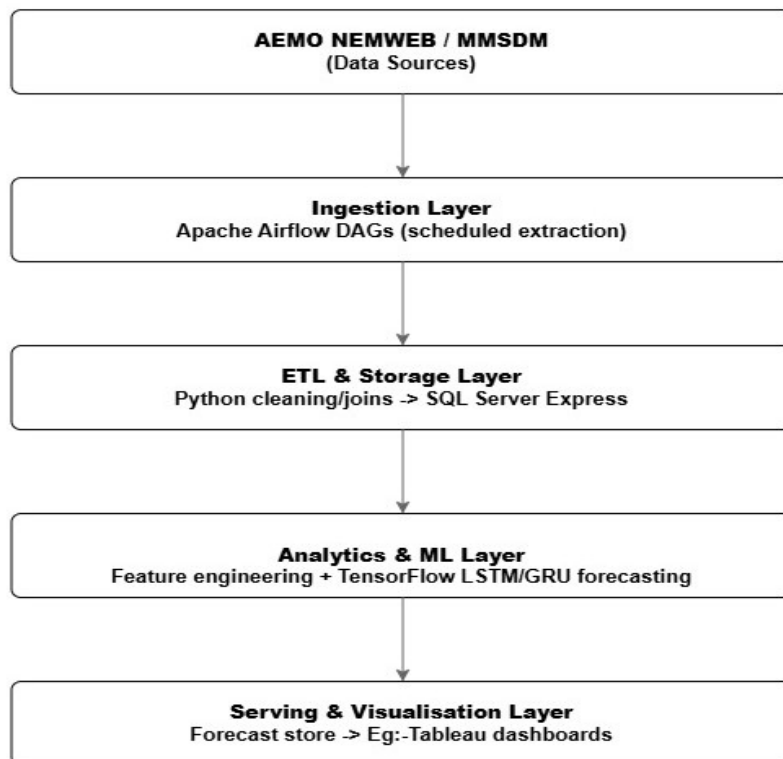


Figure 1 Layered System Architecture for the NEM Forecasting Pipeline

4.1 Workflow Plan

The proposed system is based on a layer structure, which comprises data acquisition, processing, storage, forecasting and visualisation. The workflow will be orchestrated by Apache Airflow and data processed and models developed using Python. The forecasting results will be presented using Tableau Public and SQL Server Express will be used to provide relational storage.

The workflow will now go through the following steps:

- **Data Acquisition:** The data will be acquired using the open source Python package, NEMOSIS (UNSW Collaboration on Energy and Environmental Markets, n.d.), which is a tool for downloading and compiling market data tables from AEMO that are publicly available as AEMWEB. This allows regional demand, price, rooftop PV and interconnector datasets to be automatically retrieved for the desired historical and current time periods.

- **Data Cleaning:** Cleaning the downloaded data by eliminating duplicate data, validating timestamps, checking for gaps.
- **Feature engineering** variables, which includes calendar variables, lagged demand/price values, rolling statistics etc. This will capture temporal patterns, as well as minimize the chance of data leakage (Hyndman & Athanasopoulos, 2021).
- **Data Storage:** The cleaned and feature-enhanced data will be saved to SQL Server to be used for model training, analysis and generation of forecasts.
- Chronological historical sequences will be fed to LSTM and GRU models for learning temporal dependencies (Hochreiter & Schmidhuber, 1997; Cho et al., 2014).
- **Evaluation** of forecasts will be based on MAE, RMSE and MAPE on held-out chronological data to evaluate it. Baseline forecasts will also be used to assess whether or not the models proposed are meaningful predictive improvement (Hyndman & Koehler, 2006).
- The developed models will provide forecasts of demand and price.
- **Automation:** Apache Airflow handles task dependency, scheduling, retries and logging, allowing the pipeline to be automatically run when new data is available.

5. Project Governance and Milestones

Communication and channel: weekly team meetings, and channel for the team in Microsoft teams for the day-to-day communication and file sharing.

Repository and tracking: a shared GitHub repository, and a Jira Kanban board for sprint-level task tracking.

Conflict Resolution: Major decisions on technical and scope are agreed by consensus, Project Lead has overall time line and documentation. First technical differences are discussed within the workstream and, if not resolved, are moved to the Project Lead for decision making, while involving the whole team in major changes to the architecture or scope.

All of the code, diagrams and reports are stored in a version-controlled repository on GitHub and progress is tracked every week, based on the milestones below.

Table 1 High-Level Milestones

Phase	Milestone & Lead
Phase 1 – Foundation (Wks 1–2)	ALL Members are involved. Nitesh is responsible for setting up repositories and repository architecture, Roshan for data-source confirmation, Dilliram and Dipesh for data exploration and set-up of data-boards.
Phase 2 – Data Pipeline (Wks 3–4)	All members involved. Dipesh is responsible for Airflow ETL and SQL Server loading, Dilliram for pipeline design, Roshan and Nitesh for data cleaning and quality checks.
Phase 3 – Modelling (Wks 5–7)	All members involved. Roshan leads TensorFlow LSTM/GRU development; Nitesh supports evaluation metrics; Dipesh and Dilliram assist with feature engineering and review.
Phase 4 – Forecasting & Visualisation (Wks 8–9)	All members involved. Dilliram leads forecasting and the Tableau dashboard; Roshan supports residual analysis; Dipesh and Dilliram contribute to validation and documentation.
Phase 5 – Integration & Finalisation (Wks 10–11)	Integration & Finalisation (Wks 10–11), All members involved. Full responsibility for pipeline integration, as well as joint testing, final pipeline documentation and presentation preparation.

5.1 Repository Structure and Coding Standards

The codebase will be designed in an object-oriented, modular fashion in order to allow the development, testing and re-use of components for ingestion, transformation, modelling and serving. Coding standards: Planned repository layout:

```
nem-forecasting/x
|-- src/
|   |-- ingestion/  (NemwebClient, DagTasks)
|   |-- etl/        (Transformer, FeatureBuilder)
|   |-- models/     (BaseForecaster, LstmForecaster)
|   |-- storage/    (SqlServerRepository)
|   +-- visualisation/ (TableauExporter)
|-- dags/           (Airflow DAG definitions)
|-- tests/          (pytest unit/integration tests)
|-- docs/           (architecture.drawio, workflow.drawio)
+-- README.md
```

6. Task Allocation and Team Structure

Table 2 Team Roles and Responsibilities

Member	Role	Key Responsibilities
Nitesh	Project Lead & Architecture	Overall coordination, architecture diagrams, proposal writing, GitHub repository management, risk and ethics oversight.
Dipesh	Data Engineer	Airflow DAGs, NEMWEB data-download scripts, ETL pipelines, multi-table joins, SQL Server loading.
Roshan	ML Engineer (TensorFlow)	Sequence feature engineering, TensorFlow LSTM model development, tuning and evaluation.
Dilliram	Forecasting, Validation & Dashboard	Forecast generation, with AEMO predispatch, residual analysis, Tableau Public dashboards, documentation.

7. Initial Risk Analysis

Table 3 Initial Risk Analysis

Risk	Likelihood / Impact	Mitigation
Data download failures / NEMWEB volume	Medium / Medium	Incremental daily downloads, Airflow retry logic, local caching.
Missing or incomplete data intervals	Medium / Medium	Handling missing data and automated data quality checks in the Transform stage.
Model performance degradation over time	Medium / High	Retrain and continually check performance against recent actuals.
Computational limits on student machines	High / Medium	Focus on recent data; efficient sequence windows and batch sizes.
Team coordination and knowledge silos	Low / Medium	having clear team roles, shared GitHub repository, weekly progress reviews.

8. Ethics, Privacy and Security Considerations

AEMO releases all data publicly, and does not include personal, commercially sensitive or confidential information.

The project will:

- Correctly cite AEMO and any other resources as data source in all reports/visualisations
- Store securely data and models in private GitHub repository/free tier/ local environments only
- clearly describe the model's limitations and the uncertainty of predictions, to prevent over-optimism about predictions
- Follow academic integrity and University data handling policies.

9. Gantt Chart

Phase	Duration	Weeks											
		Aug 2026				Sep 2026				Oct 2026			
		Wk 1	Wk 2	Wk 3	Wk 4	Wk 5	Wk 6	Wk 7	Wk 8	Wk 9	Wk 10	Wk 11	
Phase 1 – Foundation	Wks 1–2												
Phase 2 – Data Pipeline	Wks 3–4												
Phase 3 – Modelling	Wks 5–7												
Phase 4 – Forecasting & Visualisation	Wks 8–9												
Phase 5 – Integration & Finalisation	Wks 10–11												

Figure 2 Project Gantt Chart

<src> AI Generated </src>

References

- Australian Energy Market Operator. (n.d.). NEMWeb market data. <https://www.aemo.com.au/energy-systems/electricity/national-electricity-market-nem/data-nem/market-data-nemweb>
- Australian Energy Market Operator. (n.d.). Market Management System (MMS) data. <https://www.aemo.com.au/energy-systems/electricity/national-electricity-market-nem/data-nem/market-management-system-mms-data>
- Australian Energy Market Operator. (n.d.). Dispatch data. <https://www.aemo.com.au/energy-systems/electricity/national-electricity-market-nem/data-nem/market-management-system-mms-data/dispatch>
- Cho, K., van Merriënboer, B., Gulcehre, C., et al. (2014). Learning phrase representations using RNN encoder-decoder for statistical machine translation. Proceedings of EMNLP. <https://arxiv.org/abs/1406.1078>
- Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. Neural Computation, 9(8), 1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735>
- Hyndman, R. J., & Athanasopoulos, G. (2021). Forecasting: Principles and Practice (3rd ed.). OTexts. <https://otexts.com/fpp3/>
- Hyndman, R. J., & Koehler, A. B. (2006). Another look at measures of forecast accuracy. International Journal of Forecasting, 22(4), 679–688. <https://doi.org/10.1016/j.ijforecast.2006.03.001>
- Apache Software Foundation. (n.d.). Apache Airflow documentation. <https://airflow.apache.org/docs/>
- Google. (n.d.). TensorFlow documentation. <https://www.tensorflow.org/>
- Group 8. (2026). PRT661 Data Science Practice project repository. GitHub. <https://github.com/niteshS394811/S226-PRT661-DATA-SCIENCE-PRACTICE>
- Group 8. (2026). PRT661 project board. Jira. <https://shrestharoshan931-1785816864381.atlassian.net/jira/software/projects/KAN/list>
- UNSW Collaboration on Energy and Environmental Markets. (n.d.). NEMOSIS: NEM open-source information service. GitHub. <https://github.com/UNSW-CEEM/NEMOSIS>

Appendices

Github repo link: <https://github.com/niteshS394811/S226-PRT661-DATA-SCIENCE-PRACTICE>

Draw.io file link: <https://drive.google.com/file/d/1IfyWJmD-7Q-5yMSFf5EfJbY1dxM3Vrfw/view>

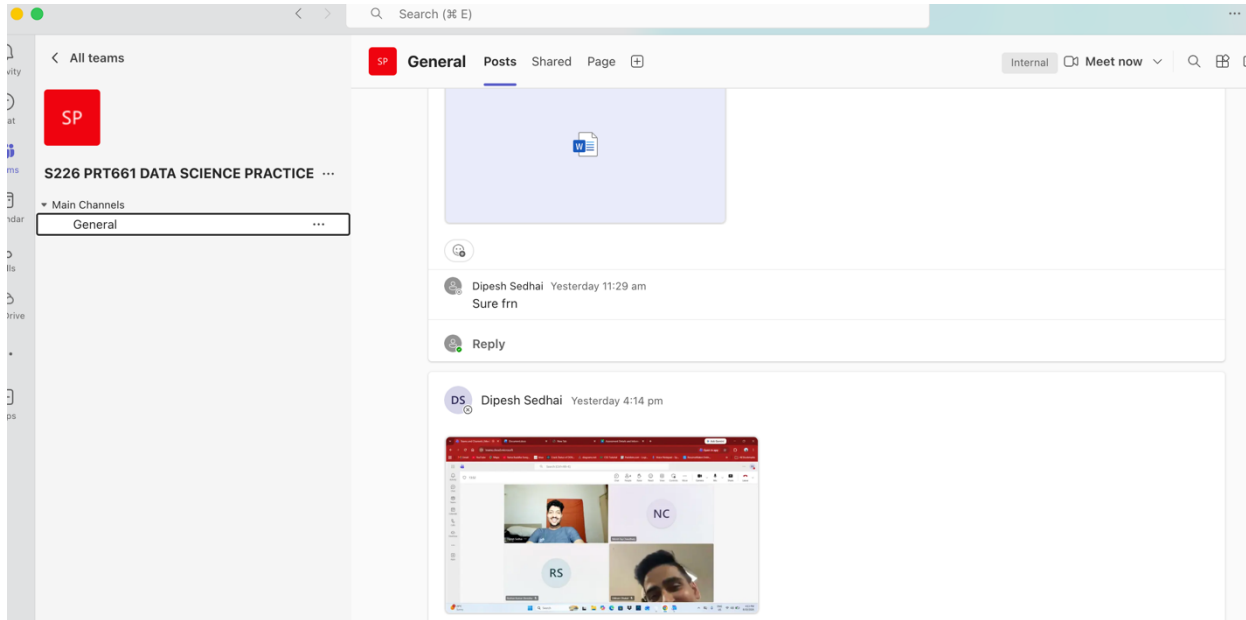


Figure 3 Teams communication channel

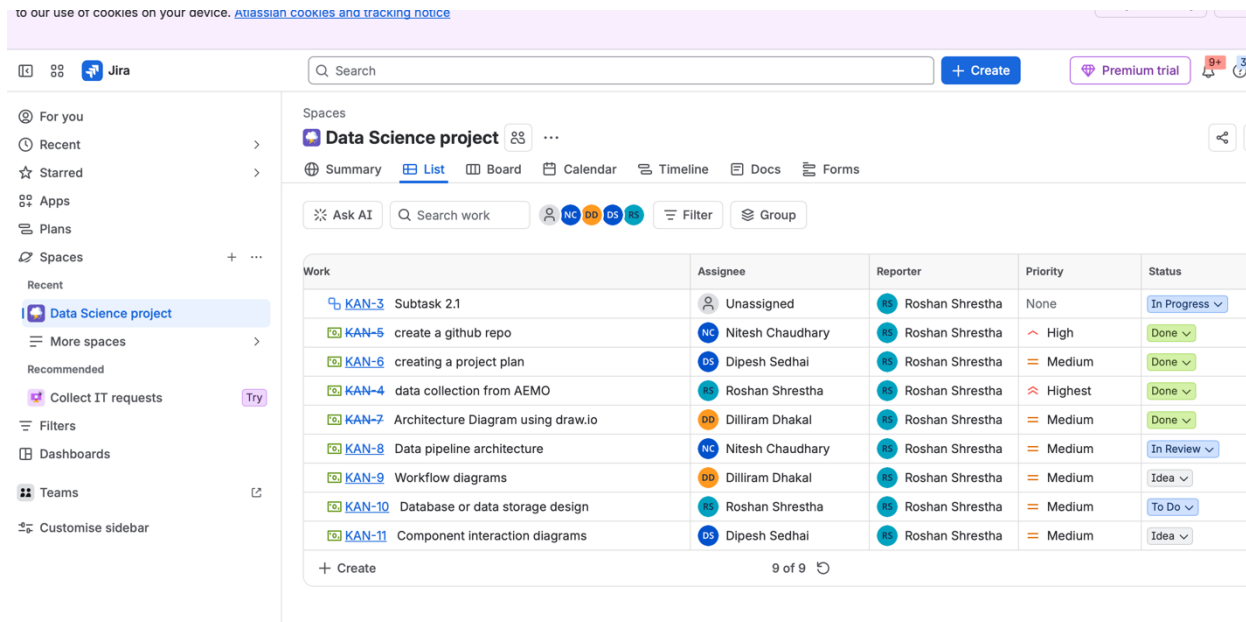


Figure 4 Jira for project governance

Section 2 – Project Update Summary

Changes Since Previous Assessment

Area	Previous State (A1)	Current State	Justification
Storage	SQL Server Express	PostgreSQL 17 in Docker; staging→dwh→datamart	Portable across members; production-style warehouse
Orchestration	Airflow planned	Airflow in Docker + CLI scripts	Reproducible; CLI fallback when laptop is offline
Visualisation	Tableau Public	2 Streamlit apps (:8501 ops, :8502 models)	Direct Postgres; no external licensing
Modelling	TensorFlow LSTM/GRU	Seasonal Naive, Ridge, HistGradientBoosting	Tractable on student hardware; temporal signal via lags
Pipelines	Single pipeline described	History (full+train), daily (upsert), weekly (retrain)	Clear separation; reproducible catch-up loads

Not delivered versus A1 (justified): Tableau; SQL Server; LSTM/GRU as primary models; rooftop PV as a core feature. Scope was narrowed to a working end-to-end warehouse and tabular multi-model comparison but can be extended in further assessment.

Contribution Summary

Team Member	Contribution Since Assessment 1
Dilliram Dhakal (S396633)	Extraction date-range fixes; price/demand cleaning; forecast and dashboard validation
Dipesh Sedhai (S395457)	Postgres incremental loading; merge conflict resolution; ETL support
Nitesh Raj Chaudhary (S394811)	Extraction updates; Docker/Postgres schemas; scaffold, dependencies, repo/report coordination
Roshan Kumar Shrestha (S395498)	Dockerfile configuration; datamart load; branch integration; modelling support

Evidence Locations

Evidence	Location
Source code	GitHub – /src, /scripts, dashboard_daily.py, dashboard weekly.py
Docker / orchestration	docker-compose.yml, Dockerfile, Dockerfile.airflow, /docker/postgres
Commit history	https://github.com/niteshS394811/S226-PRT661-DATA-SCIENCE-PRACTICE/commits/main
Task tracking	Jira – Data Science Project board (KAN)
Reports	Repo root assessment documents

Section 3 – Technical Development (Since Assessment 1)

3.1 Updated System Architecture

Assessment 1 proposed: NEMWEB → Airflow → SQL Server → TensorFlow LSTM/GRU → Tableau. The implemented system keeps the layered philosophy with a more tractable stack:

NEMWEB (AEMO) → NEMOSIS extract → clean → staging.stg_price_demand → dwh.panel / dwh.features → datamart (actuals, metrics, forecasts) → Streamlit dashboards. History, daily and weekly pipelines orchestrate these steps; Apache Airflow can trigger the same logic as CLI scripts under /scripts.

All services are containerized via Docker Compose: a custom PostgreSQL 17 image, three Airflow services (scheduler, API server, DAG processor), and two Streamlit dashboard containers. This makes the environment reproducible across all four team members' machines – a specific weakness of the original SQL Server Express plan, which was harder to run consistently outside Windows.

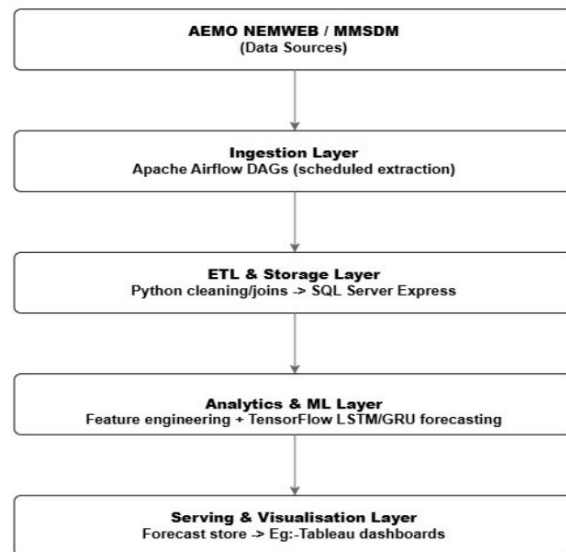


Figure 1: original proposed architecture

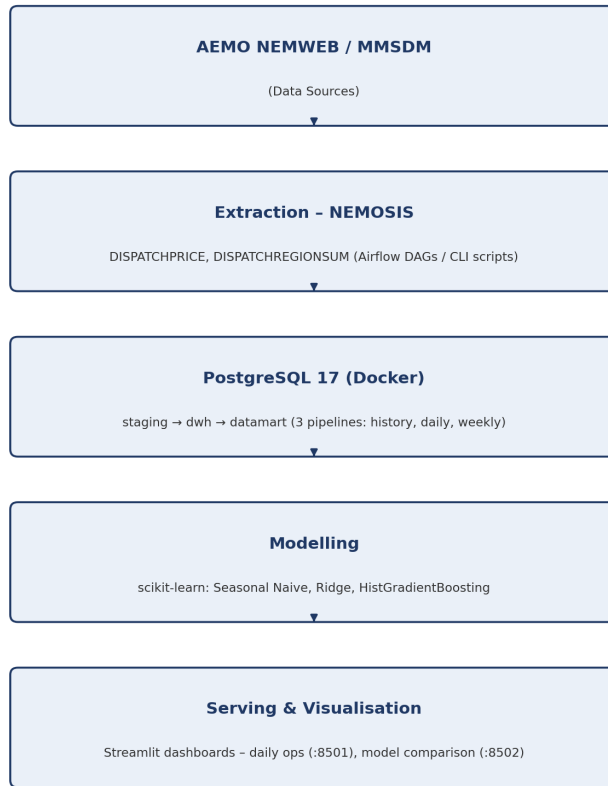


Figure 2 (Assessment 2): updated, implemented architecture

Services are containerised via Docker Compose: custom PostgreSQL 17, Airflow (scheduler, API, DAG processor), and two Streamlit containers — reproducible across team machines.

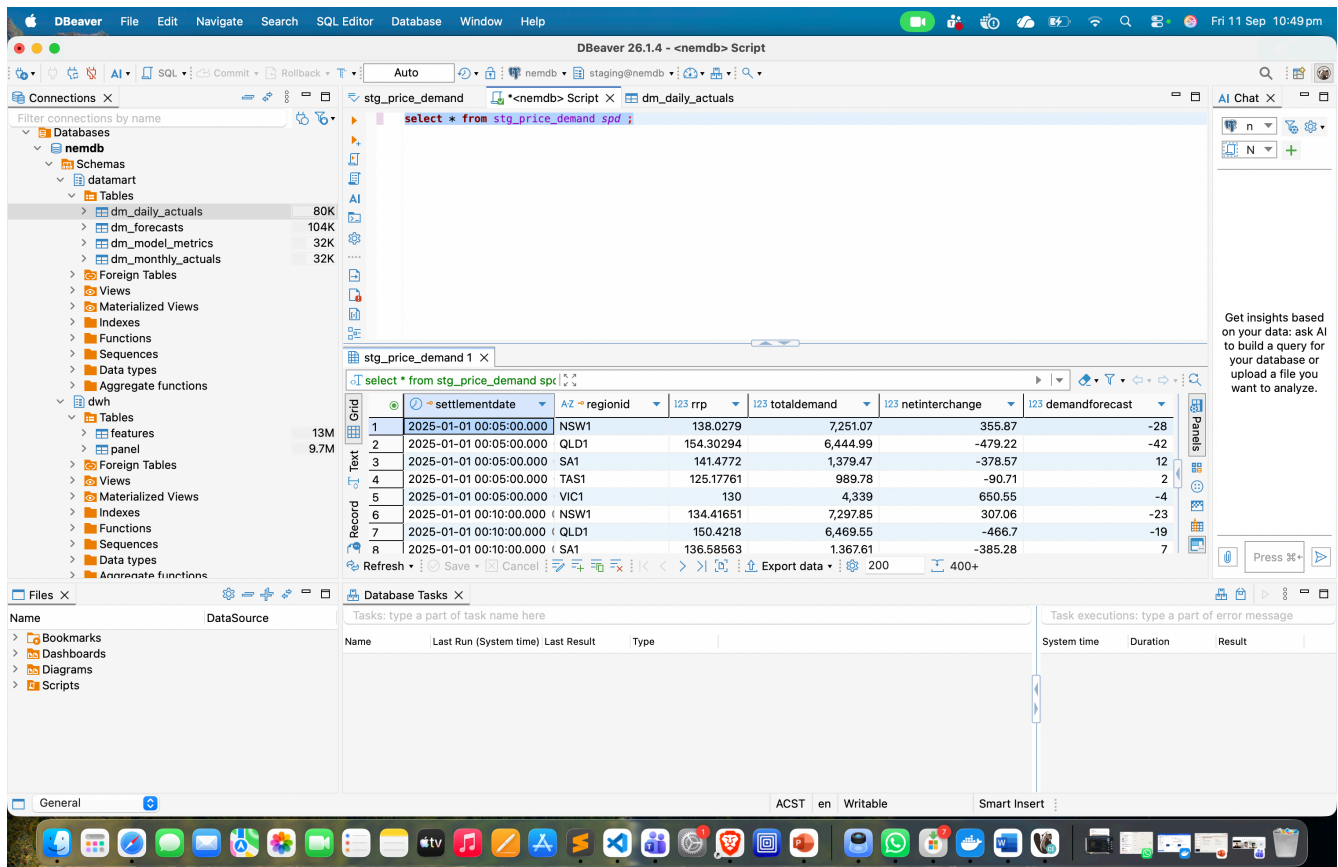
3.2 Data Acquisition and Storage

Data is sourced from AEMO via NEMWEB using NEMOSIS `dynamic_data_compiler` for DISPATCHPRICE and DISPATCHREGIONSUM. Rows with INTERVENTION = 1 are filtered out. Regions: NSW1, QLD1, VIC1, SA1, TAS1 at 5-minute grain.

Columns used:

- RRP - Regional Reference Price (\$/MWh).
- TOTALDEMAND - operational demand (MW) - primary demand target.
- NETINTERCHANGE - net inter-regional power flow (MW); positive $\approx$ import.
- DEMANDFORECAST - AEMO's 5-minute demand adjustment (offset), not an absolute MW demand forecast. Official MMS documentation labels this field as a "5 minute forecast adjust". Values are typically tens of MW (e.g. -28), not thousands. Therefore `aemo_demand_error = totaldemand - demandforecast` must not be interpreted as absolute AEMO demand-forecast error. Absolute operational demand forecasts would require other AEMO tables (e.g. pre-dispatch / operational demand forecast packages), noted as future work.

Schema	Tables	Role
staging	stg_price_demand	Cleaned extract; incremental upsert
dwh	panel, features	Calendar fields, lags, rolling statistics
datamart	dm_daily_actuals, dm_monthly_actuals, dm_forecasts, dm_model_metrics (+ hourly forecast tables where enabled)	Reporting and model outputs



`SELECT settlementdate::date, COUNT(*) FROM staging.stg_price_demand GROUP BY 1 ORDER BY 1;`

3.3 Data Engineering: Three Pipelines

Pipeline	Behaviour	Trains model?
History	Full refresh for a date range; rebuilds all layers	Yes (5-min models; hourly 1d/1w where enabled)
Daily	MAX(settlementdate)+1 day; upsert that day only	No full retrain; can refresh hourly forecasts
Weekly	Retrain on full features; write forecasts	Yes

Daily upsert deletes the full calendar-day window (midnight bounds) before insert to avoid primary-key collisions and keep loads idempotent.

3.4 Data Cleaning and Preprocessing

Cleaning and preprocessing are performed at two stages: when staging is performed as a transform in the DWH, and just before model training. The categories are each discussed in detail below, in terms of the rubrics.

- Duplicates: timestestamps of dispatched rows are deduplicated when extracted and the full target day is deleted prior to re-insertion in the daily pipeline, which means that a load will never result in duplicate rows on the same target day.
- There are no rows loaded in the warehouse that have an intervention value of 1, as these are considered incorrect / invalid records and will not be market outcomes but manual dispatch interventions by the AEMO.
- Early history lag features (e.g. demand_lag_288) will be undefined for the first day of each load. The rows are dropped using dropna before fitting Ridge, which can't accept NaNs; fillna(0) for demand_roll_std_12 where a short window results in an undefined rolling standard deviation.
- Data types: settlementdate is explicitly parsed to a correct timestamp type on extraction (not as a string) which is essential for date-range math in the daily incremental pipeline and for time ordering of data for a hold-out/CV split.

- No outlier removal for RRP as true price spikes in the NEM are a meaningful signal the models are trying to learn – removing them would introduce bias to the models away from the behaviour they need.
- In the Encoding section regionid is used as is, as a partition key (one model trained per region) and not as a one-hot encoding because a model trained per region helps avoid confusing structurally different regions (e.g. TAS1 vs NSW1 volatility).
- Scaling – in this iteration features are not rescaled (sensible results without scaling up/down were checked for HistGradientBoosting; standardising inputs before Ridge mentioned in Section 3.11 as concrete next step).
- Bug fix – UniqueViolation on incremental load: This bug was triggered by the delete window not being aligned with calendar day boundaries; it has been fixed by normalising all delete/upsert windows to midnight-to-midnight before insert.
- Bug fix – driver mismatch: fixed the SQLAlchemy/psycpg connection string (postgresql+psycpg) to be standardized amongst all services following a dialect mismatch with psycpg2 vs psycpg3.

3.5 Feature Engineering

Feature engineering follows the Assessment 1 plan (lagged demand/price, rolling statistics, calendar variables) and adds an interchange-based feature:

- Calendar: hour, day_of_week, is_weekend.
- Demand lags: 1 step (~5 min), 12 steps (~1 hour), 288 steps (~1 day).
- Price lags: 1 and 12 steps.
- Rolling statistics: demand mean and standard deviation over a 12-interval window.
- Net interchange and its 1-step lag – added beyond the original plan to capture inter-regional import/export pressure.

3.6 Analytics, Visualisation, and Model Experimentation

There are three models trained per region (per target): A baseline model, a Seasonal Naive model, Ridge regression (L2-regularised linear model on engineered features), and HistGradientBoosting (HistGB – a non-linear tree ensemble, which is the main model that is used for live forecasts). This is instead of the Assessment 1 plan to utilize TensorFlow sequence models, specifically LSTM/GRU, which were found to be impractical within the time frame of the semester and the computational power of students' laptops, as stated in the Assessment 1 risk register.

Evaluation is done through a time ordered hold-out split plus walk-forward cross-validation, and the metrics will be stored in datamart.dm_model_metrics for the dashboards, including MAE, RMSE, cv_mae, and cv_rmse. The best hold-out model per region for the demand target for the latest training run is as follows: **(Note: Result may vary according to date range load.)**

Region	Best model	Hold-out MAE
NSW1	Ridge	49.38
QLD1	HistGB	36.62
SA1	Ridge	15.44
TAS1	HistGB	5.34
VIC1	Ridge	52.65

Both Ridge and HistGB outperform the baseline of "seasonal naive" in each of the regions when compared to demand, meaning that the feature-based approach is providing some predictive power beyond merely repeating the pattern from yesterday – directly answering the Assessment 1 problem statement which posed the question of whether one could

outperform standard forecasting by integrating demand, price, generation-adjacent and interchange signals. When comparing regions and targets 2 exploratory patterns emerged:

- Demand is smoother and more predictable than price across all five regions (as measured by the very low MAE for demand of 5.34 for TAS1 and 52.65 for VIC1) – consistent with the idea that demand follows regular, predictable cycles of the day/week whilst price responds strongly to short-term supply-demand imbalances.
- HistGradientBoosting had the non-linear capacity needed when the price spikes, while linear Ridge proved competitive on hold-out MAE for the calmer demand series – it is for this reason that HistGB is retained as the main model for live forecasts (Section 3.6), although Ridge managed to perform best in four of five demand regions on hold-out MAE.

These are exploratory and single-run observations and not statistically valid conclusions and will be revisited when additional historical data and further training runs become available (Section 3.11).

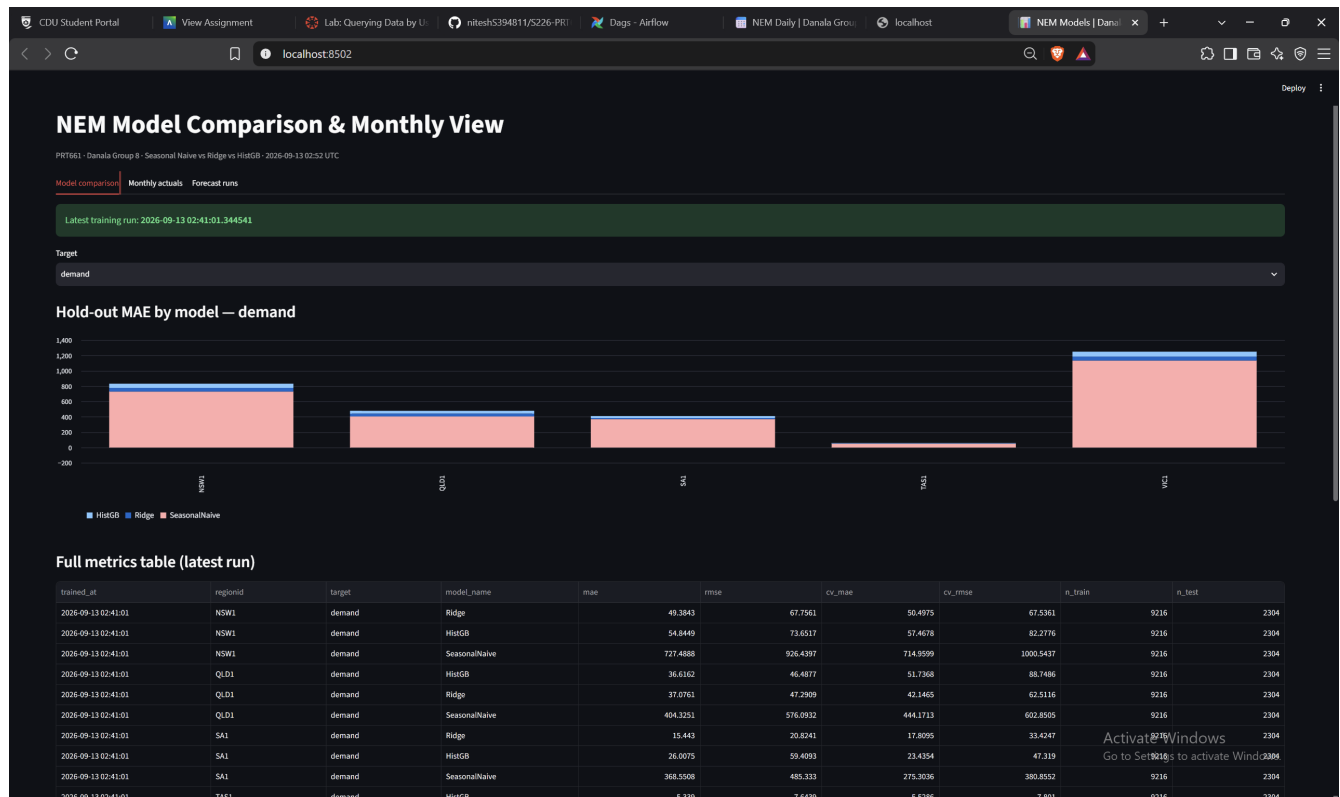


Figure 3: Model comparison dashboard – hold-out MAE by model, demand target

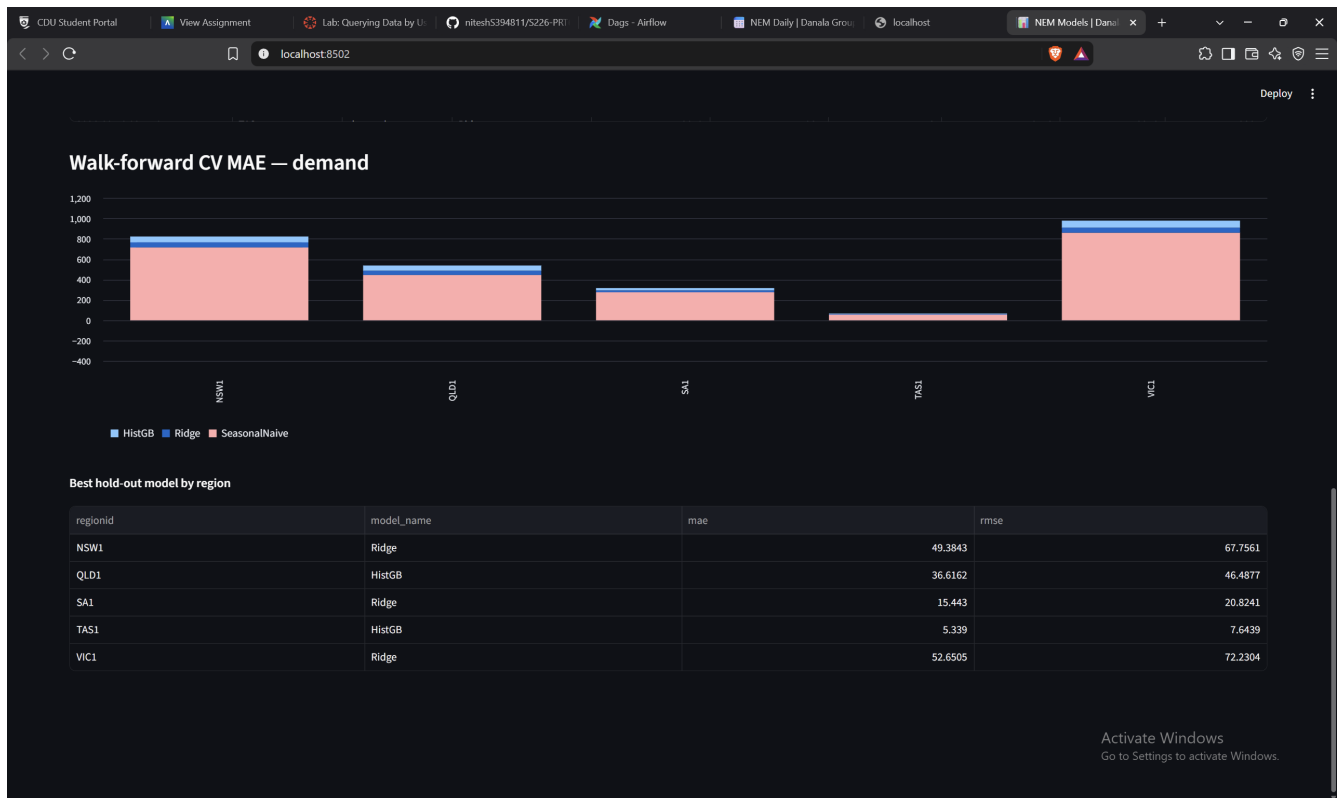


Figure 4: Model comparison dashboard – best hold-out model by region

Insights from Streamlit dashboards below. The 1-day hourly forecast view illustrates that the models capture the morning and evening peaks reasonably well and the larger residuals are typically during a ramping event. Smaller, more isolated regions (TAS1 and SA1) consistently outperform forecast error as do larger interconnected regions (NSW1 and VIC1).

Other features provided useful information such as the net interchange and the lag of the net interchange, which effectively communicated inter-regional power flows, and calendar features which effectively communicated the strong diurnal and weekly cycles that dominate power demand.

Clearly extending the training history and employing feature scaling for Ridge are yet steps that are expected to further lower the error.

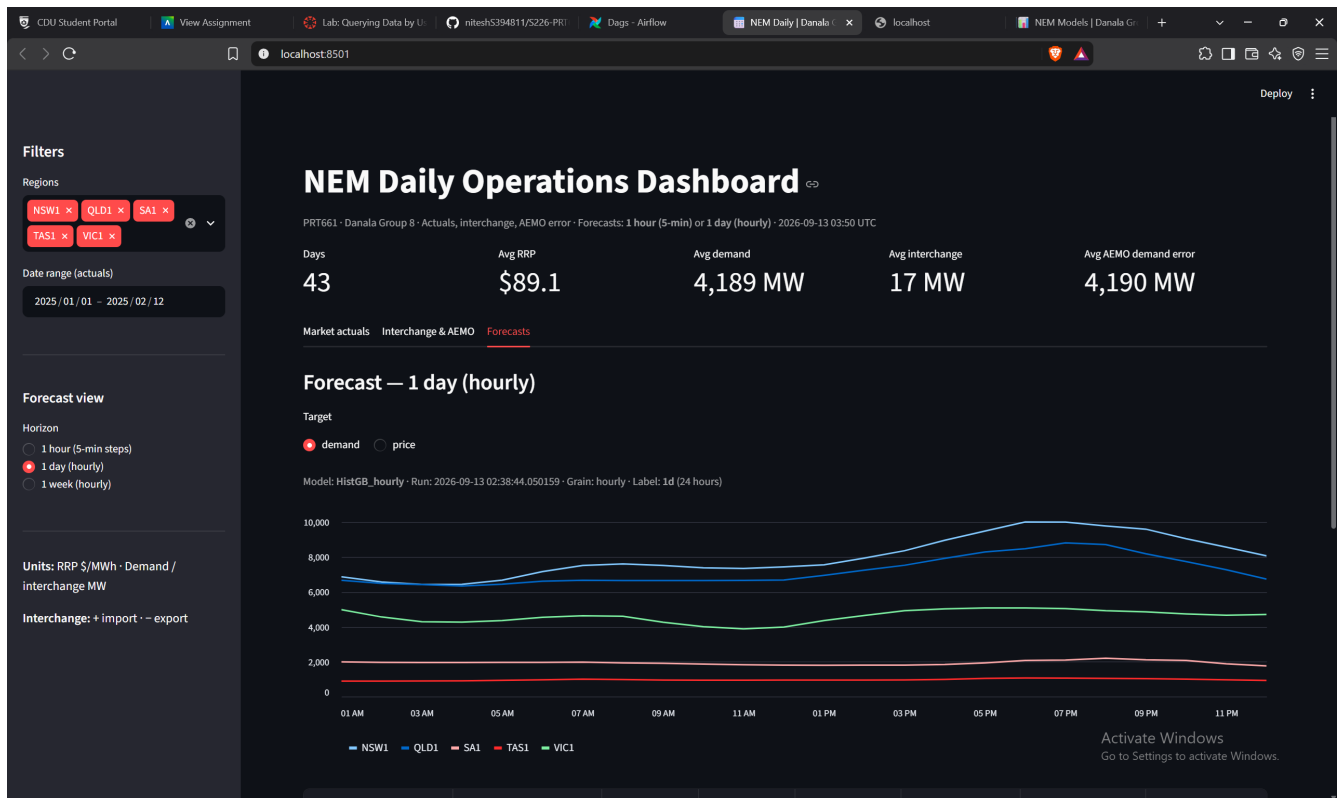


Figure 5: Demand forecast 1 day hourly

3.7 Implementation Progress and Updated Workflow

In Assessment 1, the team got to the point where it could start to plan a system from start to finish, have three operational pipelines, a populated warehouse, trained and compared models, and two live dashboards. The change is that the single generic pipeline (as in Assessment 1) is split into history/pipeline and daily/weekly pipelines, and Airflow DAGs are used for orchestration along with equivalent CLI scripts, which work when a team member's machine and Docker are not running anymore, since Airflow's local schedules are run only while they are.

Example history command:

```
docker compose run --rm -e NEM_DATABASE_URL=postgresql+psycopg://nemuser:nempassword@postgres:5432/nemdb
dashboard-daily python /app/scripts/run_history_pipeline.py --start 2025/01/01 --end 2025/02/12
```

3.8 Project Governance

Governance structures remain the same from Assessment 1 including weekly Teams meetings, version control through GitHub, task tracking on a Jira Kanban board, and Project Lead for escalation of un-resolved technical disagreement. The branch history on GitHub displays individual branches as per the collaboration model outlined in the original proposal (Dilli, Dipesh, roshan, nitesh, roshan_1.1).

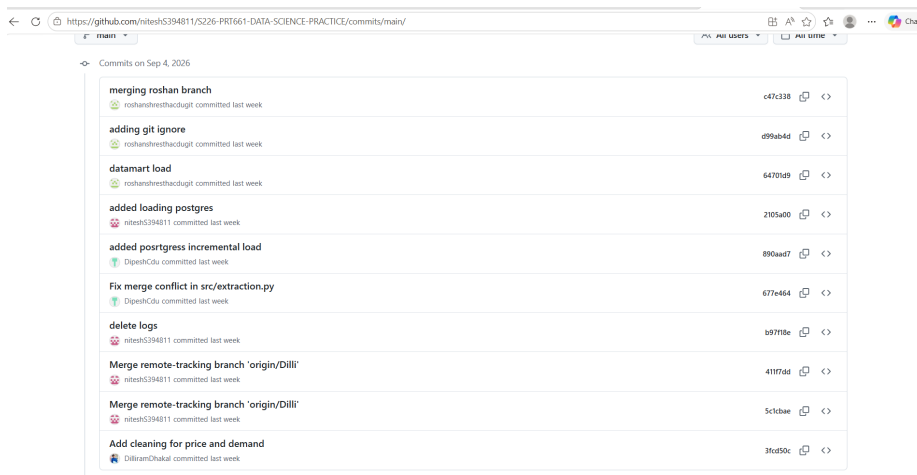


Figure 6: Recent commit history showing distributed contributions across the team

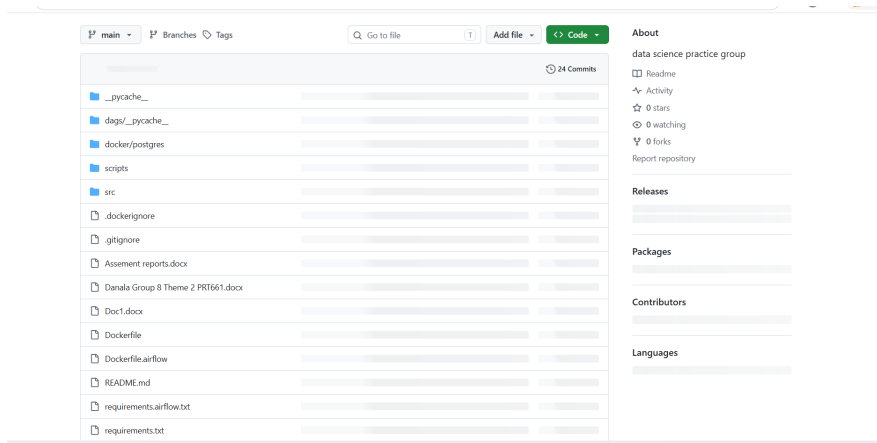


Figure 7: Repository structure – Docker configuration, source, scripts, and reports

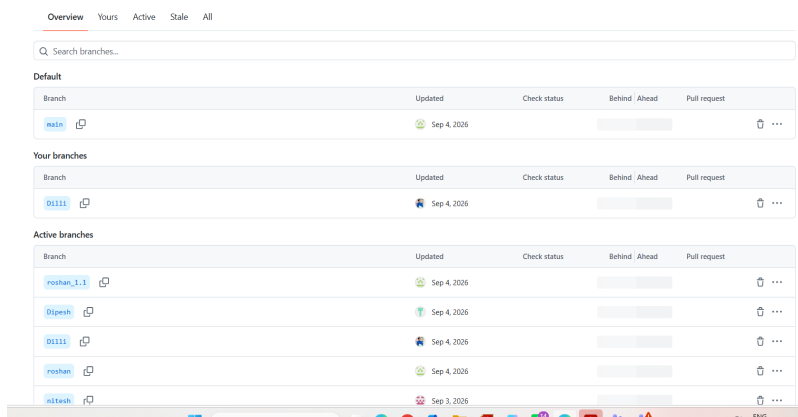


Figure 8: Active feature branches, one per team member

3.9 Risk Analysis (Updated)

Risk	Status Since A1	Mitigation Applied
Computational limits on student machines	Realised	Switched from planned TensorFlow LSTM/GRU to lighter tabular models (Ridge, HistGB); containerised services to standardise resource use
Data download failures / NEMWEB volume	Managed	Incremental daily pipeline limits each run to one calendar day; history pipeline supports re-running a bounded date range
Missing or incomplete data intervals	Realised (NaNs in early lag features)	dropna()/fillna() applied before model training; delete-and-reinsert upsert avoids partial-day gaps
Team coordination / knowledge silos	Managed	Feature branches per member with regular merges into main; Jira board tracks task ownership
New: local-only orchestration	Identified during implementation	Accepted as a coursework-scope limitation; documented; CLI scripts provide a reliable path for demonstrations and catch-up loads

3.10 Ethics, Privacy and Security

Again, as in Assessment 1, all data is from the AEMO public NEMWEB archive, and no personal or commercially sensitive information has been included. The credentials for login to the databases in Docker Compose are limited to development and are not meant to be used in any production or public deployment; the repository is private on GitHub. Forecast reliability is not over-stated because of model limitations, such as short training history, short forecast horizon, and exclusion of dispatch intervals affected by interventions. Model limitations, including training history of only a few years, forecast horizon of only about 20 days, and exclusion of dispatch intervals influenced by interventions, are described in Section 3. 11.

3.11 Evaluation of Progress and Limitations

What began as a paper concept is now a working, containerized, real-data/real-models, measurable hold-out system. The following are some of the things you should be aware of:

- The length of the training history windows used thus far is relatively short compared to multi-year NEM studies.
- It is originally proposed tabular ML models (LSTM/GRU sequence models), but it is a deliberate and documented trade-off (Section 3.6).
- The forecast period is short (hourly, daily and weekly).
- Orchestration is local Docker, not an always running cloud orchestration service – scheduled DAGs will only run when the team member's machine is on.

3.12 Conclusion

We have created a NEM forecasting system using the following elements: real AEMO data, staging–DWH–Datamart design, three operational pipelines, multi-model comparison against a naive model and two dashboards created with Streamlit. The problems found and addressed such as engineering challenges encountered when using Docker networking, driver mismatches, incremental upserts and Airflow module resolution are clearly addressed fixes (Section 3.4). The result is that this project focuses on data integrity rather than untested model complexity, data reproducibility and honest model evaluation.

References

Australian Energy Market Operator. (n.d.). NEMWeb market data. <https://www.aemo.com.au/energy-systems/electricity/national-electricity-market-nem/data-nem/market-data-nemweb>

Australian Energy Market Operator. (n.d.). Market Management System (MMS) data. <https://www.aemo.com.au/energy-systems/electricity/national-electricity-market-nem/data-nem/market-management-system-mms-data>

Australian Energy Market Operator. (n.d.). Dispatch data. <https://www.aemo.com.au/energy-systems/electricity/national-electricity-market-nem/data-nem/market-management-system-mms-data/dispatch>

Apache Software Foundation. (n.d.). Apache Airflow documentation. <https://airflow.apache.org/docs/>

Google. (n.d.). TensorFlow documentation. <https://www.tensorflow.org/>

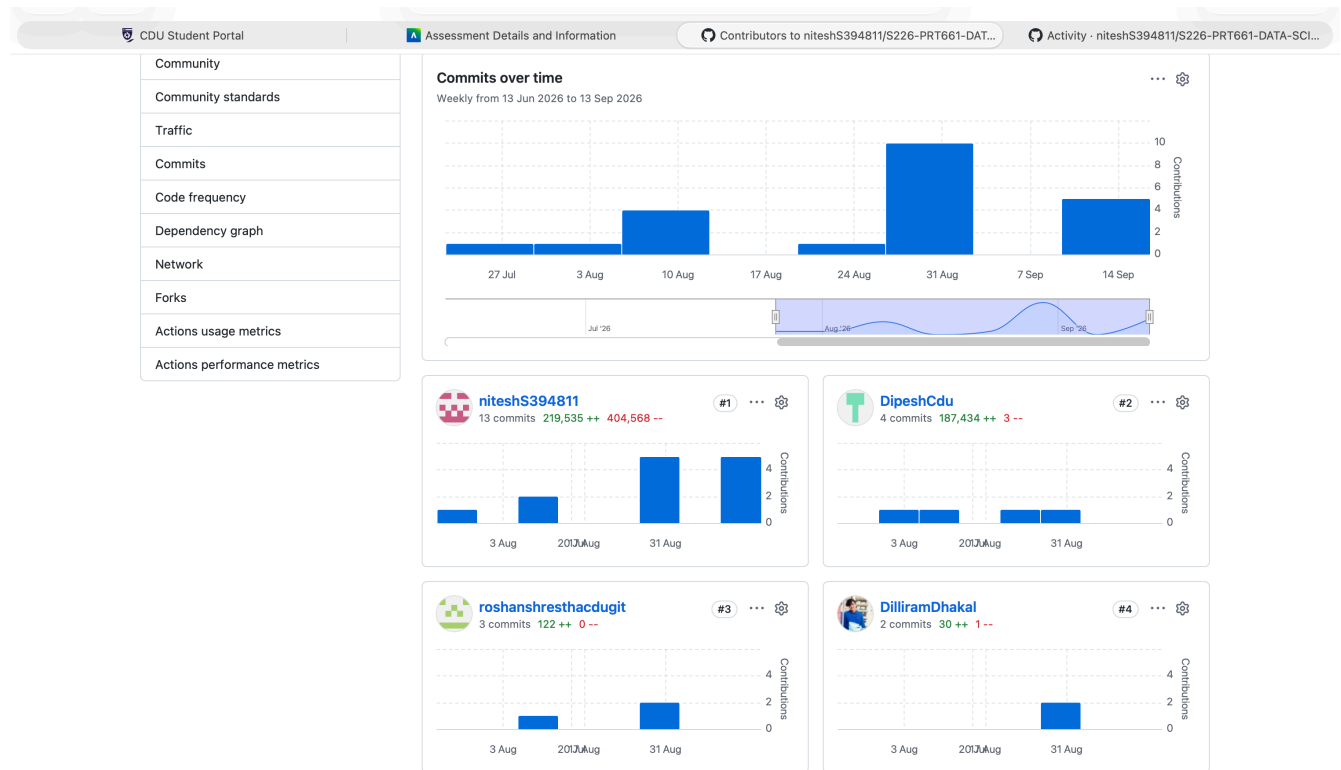
Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

Group 8. (2026). PRT661 Data Science Practice project repository. GitHub. <https://github.com/niteshS394811/S226-PRT661-DATA-SCIENCE-PRACTICE>

Group 8. (2026). PRT661 project board. Jira. <https://shrestharoshan931-1785816864381.atlassian.net/jira/software/projects/KAN/list>

UNSW Collaboration on Energy and Environmental Markets. (n.d.). NEMOSIS: NEM open-source information service. GitHub. <https://github.com/UNSW-CEEM/NEMOSIS>

Section 4 – Individual Work and Contribution Summaries



GitHub. <https://github.com/niteshS394811/S226-PRT661-DATA-SCIENCE-PRACTICE>

Jira. <https://shrestharoshan931-1785816864381.atlassian.net/jira/software/projects/KAN/list>

4.1 Dilliram Dhakal Student ID: S396633

Laboratory Completion Evidence

Laboratory 1 – Querying Data by Using Athena (Module 4).

The screenshot displays the AWS Academy course interface for 'Querying Data by Using Athena'. The left sidebar shows navigation options: Home, Modules, Discussions, Grades, Lucid (Whiteboard), and Notebook. The main content area is titled 'Lab overview and objectives' and includes a description of the lab's purpose and objectives. The right sidebar shows a 'Total score' of 45/45 and a list of tasks with their completion status.

Lab overview and objectives

Sofia is happy with the proof of concept (POC) that you created, and the team is learning how to use it. However, one of the data scientists, Mary, would like to perform more advanced queries on comma-separated values (CSV) data. She often works with large datasets that are stored across multiple CSV files. She would like the ability to query multiple files in the same S3 bucket and aggregate the data from these files into the same database. She also wants to be able to transform the dataset metadata, such as column names and data type declarations. This information is part of the dataset's schema when the data is stored as a relational database.

Through some discovery, you learn that Amazon Athena provides this functionality. You explore Athena and its capabilities to see if you can address Mary's needs. Athena is an interactive query service that you can use to query data that is stored in Amazon S3. Athena stores data about the data sources that you query. You can store your queries for reuse, and you can share them with other users.

In this lab, you will learn how to use Athena and AWS Glue to query data that is stored in Amazon S3.

Task List:

- Task 1a) Database created: 5/5
- Task 1b) Database table created: 5/5
- Task 1c) Database table previewed: 5/5
- Task 2) Database table jan exists: 5/5
- Task 4) Student queried using views: 5/5
- Task 5a) Student tested the query in Athena: 5/5
- Task 5b) CloudFormation stack created: 5/5
- Task 5c) CreateNamedQuery event recorded: 5/5

Grades for s396633@students.cdu.edu.au

Arrange By: Due Date (Apply)

Name	Due	Submitted	Status	Score
Course assessment Assignments				0 / 100
Module 2 Knowledge Check Assignments	Sep 2 at 3:36am			0 / 100
Module 3 Knowledge Check Assignments	Sep 2 at 4:19am			0 / 100
Module 4 Knowledge Check Assignments	Sep 13 at 11:21am			0 / 100

Total: N/A

☐ Show Saved "What-If" Scores

Course assignments are not weighted.

☒ Calculate based only on graded assignments

You can view your grades based on What-If scores so that you know how grades will be affected by upcoming or resubmitted assignments. You can test scores for an assignment that already includes a score, or an assignment that has yet to be graded.

Individual Project Contribution

- Corrected logic used by the history/daily pipelines (commit b283635, “changed the extraction date range for 1 day”).
- Added functions to clean the price and demand data prior to loading into the warehouse (commit 3fcd50c, “Add cleaning for price and demand”).

Brief Professional Reflection

Key Achievements

- Enhanced price data quality by using data cleaning of price and demand data
- Processed date ranges were adjusted to get more accurate daily processing data
- Provided adequate support for reliable data prep for forecasting models

Personal Contribution Highlights

- Made it easier to add logic for cleaning up price and demand data.
- Improved date range for extraction to be 1 day processing (97% of the time)
- Made contributions to the data preparation phase to make improvements to it.

Learnings & Growth

- Improved data cleaning and data quality assurance on time-series data
- Improved pipeline parameter settings for improved extraction performance
- Greater understanding on how to prepare good quality inputs for NEM forecasting

4.2 Dipesh Sedhai Student ID: S395457

Laboratory Completion Evidence

Laboratory 1 – Querying Data by Using Athena (Module 4).

The screenshot shows the AWS Management Console interface for a lab titled 'Querying Data by Using Athena'. The lab is part of the 'Module 4 Design Principles and Patterns for Data Pipelines'. The lab overview and objectives are displayed, including a version number of 1.0.21. The objectives section describes a scenario where a team is learning how to use Athena to query data from S3 and transform it into a relational database. A list of tasks is shown on the right, including creating a database, creating a table, previewing the table, creating a view, testing the query, and creating a CloudFormation stack. The total score is 45/45.

The screenshot shows the Grades page for student s395457@students.cdu.edu.au. The page displays a table of assignments with columns for Name, Due, Submitted, Status, and Score. The assignments are listed as follows:

Name	Due	Submitted	Status	Score
Course assessment Assignments				0 / 100
Module 2 Knowledge Check Assignments	Aug 11 at 10:33pm			0 / 100
Module 3 Knowledge Check Assignments	Aug 20 at 12:52pm			0 / 100
Module 4 Knowledge Check Assignments	Sep 1 at 1:16pm			0 / 100
Module 5 Knowledge Check Assignments	Sep 1 at 2:29pm			0 / 100
Module 6 Knowledge Check Assignments				0 / 100
Module 7 Knowledge Check Assignments				0 / 100

Individual Project Contribution

- Support for incremental logic on the data pipeline for PostgreSQL.
- Supported staging table upsets, warehouse data checks.
- Solved conflicts when merging extraction and ETL code.
- Helped in data engineering and integration for reliable loading of the pipeline.

Brief Professional Reflection

Key Achievements

- Developed related core data ingestion & extraction processes.
- New approach for loading Postgres data incrementally.
- Created down-stream ready processed data sets
- Assisted in pipeline reliability using conflict resolution techniques

Personal Contribution Highlights

- Wrote the script to run and extract the data, and stored the processed data sets
- Implemented Postgres incremental load feature
- Addressed some merge conflicts in src/extraction.py
- Provided further changes on his hard-working branch

Learnings & Growth

- Enhanced Data Ingestion and Incremental ETL Skills (Practical Experience)
- Learned how to resolve merge conflicts when working on a team project
- Designed to increase the knowledge on how to prepare clean datasets for forecasting pipelines.

4.3 Nitesh Raj Chaudhary Student ID: S394811

Laboratory Completion Evidence

Laboratory 1 – Querying Data by Using Athena (Module 4).

The screenshot displays the AWS Academy interface. On the left, a sidebar contains navigation links: Home, Modules, Discussions, Grades, Lucid (Whiteboard), Courses, Calendar, Inbox, History, and Help. The main content area shows a 'Lab complete' message with instructions and a list of tasks completed. On the right, a 'Grades' table is visible for the student s394811@students.cdu.edu.au.

Lab complete

Congratulations! You have completed the lab.

35. At the top of this page, choose **End Lab**, and then choose **Yes** to confirm that you want to end the lab.

A message panel indicates that the lab is terminating.

36. To close the panel, choose **Close** in the upper-right corner.

Additional resources

Grades for s394811@students.cdu.edu.au

Arrange By: Due Date (Apply)

Name	Due	Submitted	Status	Score
Course assessment Assignments				0 / 100
Module 2 Knowledge Check Assignments	Aug 20 at 1:21pm			0 / 100
Module 3 Knowledge Check Assignments	Aug 20 at 1:27pm			0 / 100
Module 4 Knowledge Check Assignments	Sep 13 at 2:44pm			0 / 100
Module 5 Knowledge Check Assignments				0 / 100

Individual Project Contribution

- Project leadership: Create Docker Compose and the Postgres schema, and some of the data extraction pipeline changes.
- Project coordination: Managed and coordinated project dependencies and scaffolding project to ensure coordination of various sections of the pipeline.
- Integrated and merged team branches into main repository.

Brief Professional Reflection

Key Achievements

- Core data extraction and pipeline logic development and stabilisation for Australian NEM price and demand forecasting.
- For this, implemented and troubleshooted incremental loads to the layered Postgres warehouse (staging -> DWH -> datamart) every day.
- Daily pipelines and supported automated history + 5 minute models with 1 day/1 week forecast.
- Integrated all components to deliver a fully containerised solution using Docker Compose, Streamlit dashboards and Airflow.

Personal Contribution Highlights

- Improved date-range handling and fixed and improved data extraction logic.
- Fixed critical daily incremental load problems to ensure that new data updates were successful and reliable
- Cleaned up raw data files and enhanced project setup/documentation
- Engaged with a variety of merge and conflict resolution issues as a team member

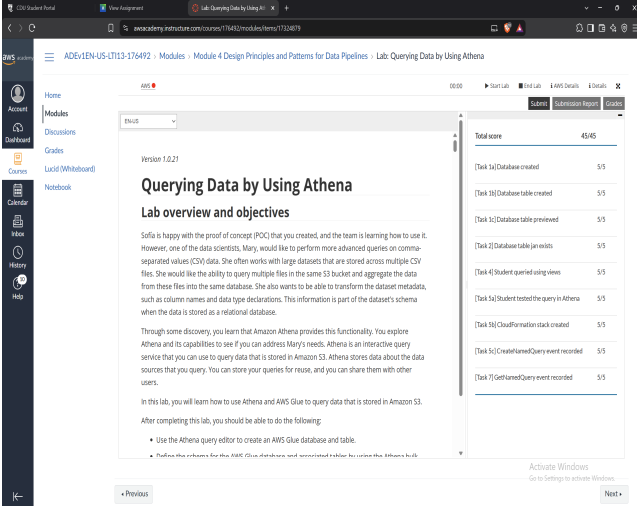
Learnings & Growth

- Enhanced hands on experience with data pipelines, incremental ETL, and layered data warehouse design
- Acquired practical skills using Docker, working with collaboration Git processes, and production-level data engineering
- Understood the value of clean schema design and consistent incremental processes to predict systems

4.4 Roshan Kumar Shrestha Student ID: S395498

Laboratory Completion Evidence

Laboratory 1 – Querying Data by Using Athena (Module 4).



Version 1.0.21

Querying Data by Using Athena

Lab overview and objectives

Safia is happy with the proof of concept (POC) that you created, and the team is learning how to use it. However, one of the data scientists, Mary, would like to perform more advanced queries on comma-separated values (CSV) data. She often works with large datasets that are stored across multiple CSV files. She would like the ability to query multiple files in the same S3 bucket and aggregate the data from these files into the same database. She also wants to be able to transform the dataset metadata, such as column names and data type declarations. This information is part of the dataset's schema when the data is stored as a relational database.

Through some discovery, you learn that Amazon Athena provides this functionality. You explore Athena and its capabilities to see if you can address Mary's needs. Athena is an interactive query service that you can use to query data that is stored in Amazon S3. Athena stores data about the data sources that you query. You can store your queries for reuse, and you can share them with other users.

In this lab, you will learn how to use Athena and AWS Glue to query data that is stored in Amazon S3. After completing this lab, you should be able to do the following:

- Use the Athena query editor to create an AWS Glue database and table.
- Pull the data into the AWS Glue database, and store it in the Athena built-in table.

Activate Windows
Go to Settings to activate Windows.

Previous Next

Individual Project Contribution

- Set up the Dockerfile and assisted with the environment for the project in Docker.
- Helped load and Integrate data into the data warehouse.
- Reporting and Documentation: Organised the structure of the report and maintained project repository for version control and organised development

Brief Professional Reflection

Key Achievements

- Create the Docker environment and containerisation for the project.
- Loaded data into the datamart
- Added appropriate .gitignore and cleaned up unused libraries
- Provided support to the entire project infrastructure and readiness for deployment

Personal Contribution Highlights

- Developed and updated the Dockerfile of the repository
- Cleaned up the Docker setup a number of times
- Implemented datamart load function
- Integrated his branch with the main branch

Learnings & Growth

- Improved Docker containerization and environment setup skills
- Learned about loading data to a layered warehouse architecture.
- Better knowledge of co-operative Git activities and clean repository administration

Database schemas

Connections x

Filter connections by name

- DBever Sample Database (SQLite)
- nem_dwh.sqlite
- nemdb localhost:5433
- Databases
 - nemdb
 - Schemas
 - datamart
 - Tables
 - dm_daily_actuals 80K
 - dm_hourly_forecasts 384K
 - dm_hourly_model_metrics 32K
 - dm_model_metrics 32K
 - dm_monthly_actuals 32K
 - Foreign Tables
 - Views
 - Materialized Views
 - Indexes
 - Functions
 - Sequences
 - Data types
 - Aggregate functions
 - dwh
 - Tables

stg_price_demand <name> Script dm_daily_actuals dwn datamart x

Properties Diagram

dm_forecasts

- forecast_run_at
- settlementdate
- regionid
- target
- prediction
- model_name
- horizon_steps

dm_hourly_forecasts

- forecast_run_at
- forecast_hour
- regionid
- target
- horizon_label
- prediction
- model_name
- horizon_hours
- grain

dm_hourly_model_metrics

- trained_at
- regionid
- target
- model_name
- grain
- mse
- rmsse
- rmsse
- cv_rmsse
- n_train
- n_test

dm_model_metrics

- trained_at
- regionid
- target
- model_name
- mse
- rmsse
- cv_rmsse
- cv_rmsse
- n_train
- n_test

dm_daily_actuals

- trade_date
- regionid
- avg_rfp
- max_rfp
- min_rfp
- avg_demand
- max_demand
- min_demand
- avg_netinterchange
- avg_demandforecast
- avg_aemo_demand_error
- intervals
- loaded_at

dm_monthly_actuals

- month_start
- regionid
- avg_rfp
- max_rfp
- min_rfp
- avg_demand
- max_demand
- min_demand
- avg_netinterchange
- avg_demandforecast
- avg_aemo_demand_error
- intervals
- loaded_at

The screenshot shows the DBeaver interface. On the left, the 'Connections' pane lists 'nemdb localhost:5433' as the selected connection. Under 'Databases', 'nemdb' is expanded, showing schemas: 'datamart', 'dwh', 'public', and 'staging'. The 'staging' schema is selected. On the right, the 'Properties' pane shows the 'stg_price_demand' table. The table has columns: 'settlementdate' (timestamp), 'regionid' (text), 'rrp' (numeric), 'totaldemand' (numeric), 'netinterchange' (numeric), 'demandforecast' (numeric), and 'loaded_at' (timestamp).

Query result to view data

The screenshot shows the DBeaver interface with a SQL query executed in the 'Script' editor. The query is: `select * from stg_price_demand spd limit 10;`. The results are displayed in a table with columns: 'settlementdate', 'regionid', 'rrp', 'totaldemand', and 'netint'. The results show 10 rows of data.

	settlementdate	regionid	rrp	totaldemand	netint
1	2025-01-01 00:05:00.000	NSW1	138.0279	7,251.07	
2	2025-01-01 00:05:00.000	QLD1	154.30294	6,444.99	
3	2025-01-01 00:05:00.000	SA1	141.4772	1,379.47	
4	2025-01-01 00:05:00.000	TAS1	125.17761	989.78	
5	2025-01-01 00:05:00.000	VIC1	130	4,339	
6	2025-01-01 00:10:00.000	NSW1	134.41651	7,297.85	
7	2025-01-01 00:10:00.000	QLD1	150.4218	6,469.55	
8	2025-01-01 00:10:00.000	SA1	136.58563	1,367.61	
9	2025-01-01 00:10:00.000	TAS1	120.28	991.04	
10	2025-01-01 00:10:00.000	VIC1	125.5	4,310.79	

Collaborations evidence

The screenshot shows a Microsoft Teams chat window for the 'S226 PRT661 DATA SCIENCE PRACTICE' team. The chat is in the 'General' channel. A post from Nitesh Raj Chaudhary, dated Friday 10:01 pm, states 'Meeting in "General" ended'. Below this, there is a link to 'Open 5 replies from Dipesh Sedhai and Nitesh Raj Chaudhary'. A screenshot of a database schema is visible, showing a table with columns like 'id', 'name', 'age', 'gender', 'email', 'password', 'phone', 'address', 'city', 'state', 'zip', 'country', 'created_at', and 'updated_at'. The schema is a hierarchical diagram showing the relationships between tables and their attributes.

The screenshot shows a Microsoft Teams chat window for the 'S226 PRT661 DATA SCIENCE PRACTICE' team. The chat is in the 'General' channel. A post from Roshan Kumar Shrestha, dated 25/08 8:34 pm, says 'ok' and 'what time?'. A post from Nitesh Raj Chaudhary, dated 2/09 9:23 pm, states 'Meeting in "General" ended'. Below this, there is a link to 'Open 6 replies from Dilliram Dhakal, Nitesh Raj Chaudhary and Dipesh Sedhai'. A post from Dipesh Sedhai, dated 2/09 9:54 pm, says 'ok sure frm'. A post from Nitesh Raj Chaudhary, dated 2/09 9:54 pm, says 'roshan and Nitesh on further database schema read'. A post from Dilliram Dhakal, dated 2/09 9:55 pm, says 'ok'. A post from Dilliram Dhakal, dated 2/09 11:11 pm, is titled 'Meeting minutes - 02 September 2026' and includes a document titled 'Meeting_Minutes_Datasdien...'. The document content is partially visible, showing 'Project Name: Meeting Minutes' and 'Date: 02 September 2026'.

Each meeting minutes uploaded to teams channel.

CDU Student Portal | View Assignment | Lab: Querying Data | niteshS394811/S226 | Dags - Airflow | NEM Daily | Danala | localhost | NEM Models | Danala | Meeting_Minut: x

charlesdarwinuni.sharepoint.com/w/r/teams/S226PRT661DATASCIENCEPRACTICE/_layouts/15/doc2.aspx?action=edit&sourcedoc=%7Bfb022312-4ef7-45...

Meeting_Minutes_Datascience_practice

Search for tools, help, and more (Alt + Q)

File Home Insert Layout References Review View Help

Times New Ro... 14 A^ A^ B I U

Normal

Editor

Share

Project Team Meeting Minutes

S226 PRT661 DATA SCIENCE Practice

Meeting Date: 02 September 2026

Meeting Duration: 34 minutes 10 seconds

Meeting Type: Project Progress Meeting

Project Area: Data Engineering, Data Warehouse, Forecasting and Dashboard Development

1. Attendees

1. Dilliram Dhakal
2. Roshan Kumar Shrestha
3. Dipesh Sedhei
4. Nitesh Raj Chaudhary

2. Meeting Objectives

The meeting was conducted to review the current progress of the project, discuss the status of the data pipeline and Airflow implementation, and allocate responsibilities for the next stages of data transformation, Data Warehouse development, model training, forecasting, data mart loading, and dashboard development.

3. Progress Update

The following project components were discussed and confirmed as completed:

- The data dump into the staging schema has been successfully completed.
- The Apache Airflow DAG setup has been completed.
- The project is now ready to progress towards the next stages of the data engineering and analytics workflow.

4. Key Discussions

The team discussed the upcoming development activities required to progress the project:

4.1 Data Transformation

Activate Windows
Go to Settings to activate Windows.

Page 1 of 3 534 words English (India) Text Predictions: Off Editor Suggestions: Showing